

CODE	COURSE NAME	CATEGORY	L	T	P	CREDIT
222ECS017	SOFT COMPUTING	PROGRAMME ELECTIVE 4	3	0	0	3

Preamble: This course introduces the salient features and techniques used in soft computing. This course covers various neural networks models, fuzzy systems, genetic algorithm concepts and their applications. On completion of the course, students will be able to acquire key skills required for solving real-life problems using soft computing techniques.

Course Outcomes: After the completion of the course the student will be able to

CO 1	Analyse soft computing techniques and their applications (Cognitive knowledge level: Analyze)
CO 2	Make use of various neural network models in problem solving (Cognitive knowledge level: Apply)
CO 3	Examine various fuzzy concepts (Cognitive knowledge level: Analyse)
CO 4	Illustrate use of fuzzy models in various applications (Cognitive knowledge level: Apply)
CO 5	Analyse genetic algorithms and their applications (Cognitive knowledge level: Analyze)
CO 6	Design, develop, implement and present solutions to simple real-life problems with popular open source library using soft computing techniques (Cognitive knowledge level: Apply)

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

PO1: An ability to independently carry out research/investigation and development work in engineering and allied streams

PO2: An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

PO3: An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program

PO4: An ability to apply stream knowledge to design or develop solutions for real world problems by following the standards

PO5: An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tool to model, analyse and solve practical engineering problems.

PO6: An ability to engage in life-long learning for the design and development related to the stream related problems taking into consideration sustainability, societal, ethical and environmental aspects

PO7: An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1			☑	☑		☑	
CO 2	☑		☑	☑	☑	☑	
CO 3			☑	☑		☑	
CO 4	☑		☑	☑		☑	
CO 5	☑		☑	☑		☑	
CO 6	☑	☑	☑	☑	☑	☑	☑

Assessment Pattern

Bloom's Category	End Semester Examination
Apply	60
Analyse	40
Evaluate	Can be evaluated using Assignments/ projects
Create	Can be evaluated using Assignments/ projects

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation Pattern:

Evaluation shall only be based on application, analysis or design based questions (for both internal and end semester examinations).

Continuous Internal Evaluation: 40 marks

- i. Preparing a review article based on peer reviewed original publications (minimum 10 publications shall be referred) : 15 marks
- ii. Course based task / Seminar/ Data collection and interpretation : 15 marks
- iii. Test paper (1 number) : 10 marks

Test paper shall include minimum 80% of the syllabus.

Course based task/test paper questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students.

End Semester Examination Pattern:

The end semester examination will be conducted by the respective College.

There will be two parts; Part A and Part B.

Part A will contain 5 numerical/short answer questions with 1 question from each module, having 5 marks for each question. Students should answer all questions. Part B will contain 7 questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem solving and quantitative evaluation), with minimum one question from each module of which student should answer any five. Each question can carry 7 marks

Total duration of the examination will be 150 minutes.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly.

For example if the average end semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is $40+20 = 60\%$.

Course Level Assessment Questions**Course Outcome 1 (CO1):**

1. Compare linear and non-linear activation functions with an example for each.
2. Show that a single layer perceptron cannot simulate a simple exclusive-OR function and give the reason.
3. Implement logical NOR function using neural network model

Course Outcome 2 (CO2)

1. Design a suitable neural network for human face recognition.
2. Illustrate with an example how Supervised Learning technique can be used to solve a multilabel classification problem
3. Compare sigmoidal and step activation functions.

Course Outcome 3(CO3):

1. Two fuzzy sets A and B defined on the English alphabets (F,E,X,Y,I,T) are defined as below:

$$A = \{(F, 0.2), (E, 0.9), (X, 0.4), (Y, 0.3), (I, 0.8), (T, 0.5)\}$$

$$B = \{(F, 0.4), (E, 0.6), (X, 0.7), (Y, 0.2), (I, 0.8), (T, 0.6)\}$$

Find the following.

- i) $A \cap B$ ii) $A \cup B^c$ iii) Verify Demorgan's Law $(A \cup B)^c = A^c \cap B^c$

2. Consider a set $P = \{P1, P2, P3, P4\}$ of four varieties of paddy plants, set $D = \{D1, D2, D3, D4\}$ of the various diseases affecting the plants and $S = \{S1, S2, S3, S4\}$ be the common symptoms of the diseases. Let R be a relation on $P \times D$ and S be a relation on $D \times S$.

	<i>D1</i>	<i>D2</i>	<i>D3</i>	<i>D4</i>		<i>S1</i>	<i>S2</i>	<i>S3</i>	<i>S4</i>		
R=	<i>P1</i>	0.4	0.2	0.9	0.5	S=	<i>D1</i>	0.2	0.2	0.7	0.9
	<i>P2</i>	0.8	0.2	0.2	0.6		<i>D2</i>	1	0	0.4	0.5
	<i>P3</i>	0.6	0.6	0.7	0.9		<i>D3</i>	0.3	1	0.5	0.8
	<i>P4</i>	0.3	0.5	0.4	0.4		<i>D4</i>	0.6	0	0.7	0.4

Obtain the association of the plants with the different symptoms of the diseases using max – min composition.

3. Fuzzy binary relation R is defined on set $A = \{1, 2, 3, 4, 5\}$ and $B = \{90, 91, 92, 93, 94, 95\}$ and represents the relation “a is much smaller than b”. It is denoted by the membership function

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$$R(a,b)= \quad \text{for } a \leq b$$

$$1 - \frac{a}{b}$$

$$0 \quad \text{Otherwise}$$

Where $a \in A$ and $b \in B$. Find the domain and range of R.

Course Outcome 4 (CO4):

1. Design a fuzzy controller to determine the wash time of a washing machine. Assume the input is dirt and grease. Use three descriptors for input variable and five descriptors for output variable. Derive the set of rules for controller action and defuzzification. Show that if the dirt are more , the wash time will be more and vice versa.
2. Illustrate the multiattribute decision making with an example.
3. Demonstrate problem solving using Mamdani fuzzy model, with an example.

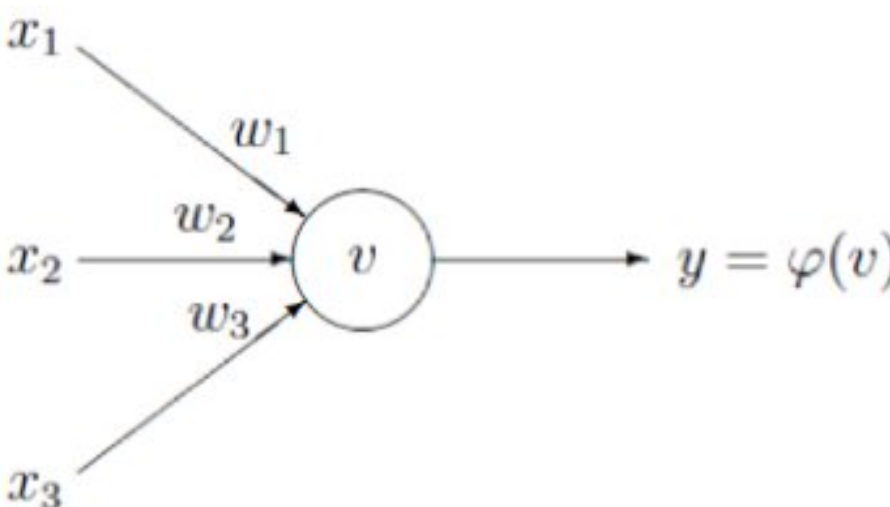
Course Outcome 5 (CO5):

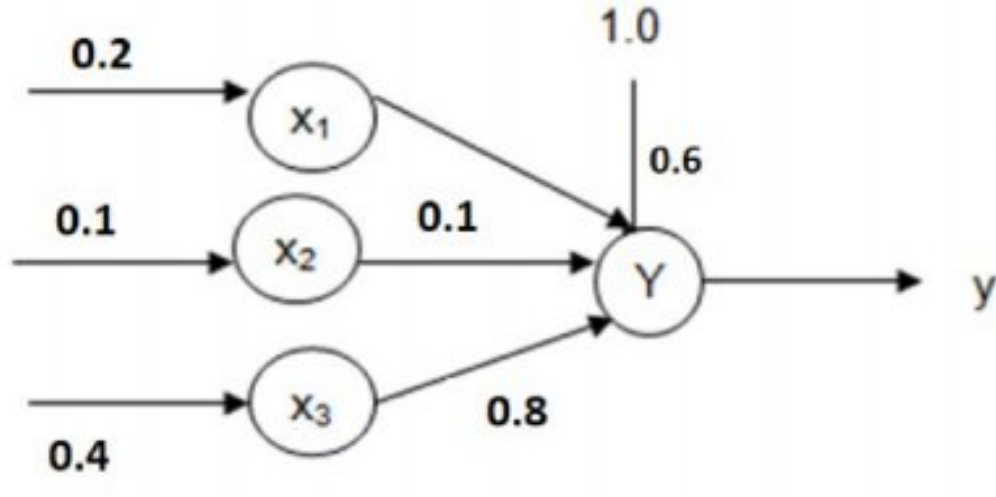
1. Illustrate Roulette Wheel Selection technique with an example.
2. Differentiate single-point crossover, multipoint crossover and uniform crossover techniques used in Genetic Algorithm.

Course Outcome 6 (CO6):

1. Prepare a review article based on peer reviewed original publications (referring minimum 10 publications) about recent neuro-fuzzy systems based applications.
2. Course project to design and develop handwritten character recognition system using Python/ TensorFlow/ Keras based on MNIST handwritten digit dataset.
3. Present a seminar on any one application of Genetic Algorithm (Other than Traveling Salesman Problem)

Model Question Paper			:COMPUTER SCIENCE AND ENGINEERING-CS2		
QP CODE:					
Reg No: _____					
Name: _____				PAGES : 3	
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY					
SECOND SEMESTER M.TECH DEGREE EXAMINATION, MONTH & YEAR					
Course Code: 222ECS017					
Course Name: Soft Computing					
Max. Marks : 60			Duration: 2.5 Hours		
PART A					
Answer All Questions. Each Question Carries 5 Marks					
1.	Implement the following truth table using suitable neural network model.				
	Input		Output		
	X1	X2			
	0	0	0		
	0	1	1		
	1	0	1		
	1	1	1		
2.	Differentiate Supervised and Unsupervised Learning Neural Networks with an example for each.				
3.	Differentiate Generalised Bell membership function and Left-Right Membership Function.				
4.	Discuss Fuzzy Expert System with an example				
5.	Explain the steps in Genetic Algorithm, with an example.				(5x5=25)
Part B					
(Answer any five questions. Each question carries 7 marks)					
6.	(a)	A handwritten character recognition system for English language has to be designed. Use feedforward neural network system to design the same.			(4)
	(b)	Implement logical X-OR function using neural network model			(3)
7.	(a)	A company has collected large amount of data in pairs of input output vectors. Design a system using radial basis function network for predicting output for new input.			(4)

	(b) Consider the unit shown in the following figure.  Suppose that the weights corresponding to the three inputs have the following values: $w_1= 1$, $w_2= - 3$, $w_3= 2$ and the activation of the unit is given by the step-function: $\varphi(v) = \begin{cases} 1 & \text{if } v \geq 0 \\ 0 & \text{otherwise} \end{cases}$ Calculate what will be the output value y of the unit for each of the following input patterns: <table><tr><th>Pattern</th><th>P_1</th><th>P_2</th><th>P_3</th><th>P_4</th></tr><tr><td>x_1</td><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>x_2</td><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>x_3</td><td>0</td><td>1</td><td>1</td><td>1</td></tr></table>	Pattern	P_1	P_2	P_3	P_4	x_1	1	0	1	1	x_2	0	1	0	1	x_3	0	1	1	1	(3)
Pattern	P_1	P_2	P_3	P_4																		
x_1	1	0	1	1																		
x_2	0	1	0	1																		
x_3	0	1	1	1																		
8.	(a) Let P and Q be two fuzzy relations defined on the sets $X \times Y$ and $Y \times Z$ respectively,where $X=\{x_1, x_2\}$, $Y=\{y_1, y_2, y_3\}$, and $Z=\{z_1, z_2\}$. P and Q are given using the following relation matrices. $P = \begin{bmatrix} 0.5 & 0.2 & 0.3 & 0.1 & 0.7 & 0.6 \end{bmatrix}$ $Q = \begin{bmatrix} 0.5 & 0.8 & 0.4 & 0.6 & 1 \end{bmatrix}$ Find the association $P \circ Q$ using both max-min composition and max-product compositions	(4)																				
	(b) Differentiate Triangular and trapezoidal fuzzy membership functions.	(3)																				

9.	(a)	Consider a two input – one output problem that includes following three rules. Rule: 1 - IF x is A3 OR y is B1 THEN z is C1 Rule: 2 - IF x is A2 AND y is B2 THEN z is C2 Rule: 3 - IF x is A1 THEN z is C3 Assume $A3=0.2$, $B1=0.4$, $A2=0.7$, $B2=0.9$, $A1=0.3$ and the selected values of z in $c1=(0,11,21)$, $C2=(40,50,60, 70)$ & $C3=(60, 70,80,90)$. Find out the value of z using Mamdani fuzzy inference system.	(5)
	(b)	Derive the conclusion for Multiple Rules with multiple antecedents.	(2)
10.	(a)	Illustrate the problem-solving method using Sugeno fuzzy system, with an example.	(4)
	(b)	A fuzzy reasoning system is provided with the following facts and rules.)Premise 1 (fact):x is A' and y is B')Premise 2 (Rule 1): if x is A and y is B , then z is C. Demonstrate the inference procedure to find out the conclusion z is C'	(3)
11.	(a)	Calculate the output of the neuron y for the following network using Binary Sigmoidal activation function. 	(4)
	(b)	Implement logical AND function using neural networks.	(3)
12.	(a)	A salesman has to follow the shortest route to visit N cities exactly once and reach the starting city. Apply genetic algorithm to solve this problem.	(4)
	(b)	Illustrate Tournament Selection in Genetic Algorithm with an example.	(3)

Syllabus

:COMPUTER SCIENCE AND ENGINEERING-CS2

Module 1 - Introduction

Soft Computing Constituents, Computational Intelligence and Soft Computing vs Artificial Intelligence and Hard Computing. Introduction to artificial neural networks- biological neurons, Mc Culloch and Pitts models of neurons, Perception networks, Multilayer Perceptron. Types of activation function, network architectures, learning process, Learning XOR and other logical Gate functions, Sigmoid Neurons, Gradient Descent Algorithm, Back propagation Neural Network.

Module 2 – Neural Networks

Adaptive Networks, Feed Forward Networks, Supervised Learning Neural Networks, Radial Basis Function Networks, Reinforcement Learning, Unsupervised Learning Neural Networks, Adaptive Resonance Architectures, BAM, Maxnet, Kohonen Self Organizing Maps, K-means clustering algorithm, Introduction to Convolutional Neural Networks, Applications of ANNs to solve real life problems. Familiarisation of neural network tools , Building neural networks with Python, TensorFlow, Introduction to Keras.

Module 3 – Fuzzy Sets & Logic

Fuzzy versus Crisp, Fuzzy sets, Membership function, linguistic variable, basic operators, properties, Extension principle and Fuzzy relations, Cartesian product, Operations on Fuzzy sets , Operations on fuzzy relations, Fuzzy Logic

Module 4 – Fuzzy Reasoning

Fuzzy If-Then Rules, Fuzzy Inference Systems, Fuzzy Expert Systems, Fuzzification and Defuzzification methods, Fuzzy Decision Making , Mamdani Fuzzy Models, Sugeno Fuzzy Models, Applications of Fuzzy logic, Neuro Fuzzy Systems.

Module 5 – Genetic Algorithm

Introduction to genetic algorithm and hybrid systems - Genetic algorithms , natural evolution , properties, classification, GA features, coding, selection, selection methods, reproduction , cross over and mutation operators, basic GA and structure, Solving Travelling Salesman Problem using GA. Applications using GA, Hybrid Systems - GA based BPNN.

Course Plan

No	Topic	No. of Lectures (36 Hours)
1	Module 1 (Introduction to Soft Computing And Neural Networks) -7 Hours	
1.1	Soft Computing Constituents, Computational Intelligence and Soft Computing vs Artificial Intelligence and Hard Computing. Introduction to artificial neural networks, biological neurons	1
1.2	Mc Culloch and Pitts models of neurons, Perception networks	1
1.3	Multilayer Perceptron, Types of activation function	1
1.4	Network architectures, Learning process	1
1.5	Learning XOR and other logical Gate functions	1
1.6	Sigmoid Neurons, Gradient Descent Algorithm	1
1.7	Back propagation Neural Network.	1
2	Module 2 (Neural Networks) 8 Hours	
2.1	Adaptive Networks, Feed Forward Networks, Supervised Learning Neural Networks	1
2.2	Radial Basis Function Networks, Reinforcement Learning,	1
2.3	Unsupervised Learning Neural Networks, Adaptive Resonance Architectures,	1
2.4	BAM, Maxnet, Kohonen Self Organizing Maps,	1
2.5	K-means clustering algorithm, Introduction to Convolutional Neural Networks,	1
2.6	Applications of ANNs to solve real life problems. Familiarisation of neural network tools	1
2.7	Building neural networks with Python	1
2.8	TensorFlow, Introduction to Keras.	1

3	Module 3 (Fuzzy Sets & Logic) 7 Hours	
3.1	Fuzzy versus Crisp, Fuzzy sets, <u>Membership function</u> , linguistic variable	1
3.2	Basic operators, properties	1
3.3	Fuzzy relations <u>max min, max product</u>	1
3.4	Cartesian product	1
3.5	Operations on Fuzzy sets	1
3.6	Operations on fuzzy relations	1
3.7	Fuzzy Logic	1
4	Module 4 (Fuzzy Reasoning) 7 Hours	
4.1	Fuzzy If-Then Rules	1
4.2	Fuzzy Inference Systems, Fuzzy Expert Systems	1
4.3	Fuzzification and <u>Defuzzification methods</u>	1
4.4	Fuzzy Decision Making	1
4.5	<u>Mamdani Fuzzy Models</u>	1
4.6	<u>Sugeno Fuzzy Models</u>	1
4.7	Applications of Fuzzy logic, Neuro Fuzzy Systems.	1
5	Module 5 (Genetic Algorithm) 7 Hours	
5.1	Introduction to genetic algorithm and hybrid systems	1
5.2	Genetic algorithms , Natural evolution , Properties	1
5.3	Classification, GA features, Coding, Selection, Selection methods	1
5.4	Reproduction - Cross over and Mutation operators	1
5.5	<u>Basic GA and structure</u> , Solving Travelling Salesman Problem using GA	1
5.6	Applications using GA	1
5.7	Hybrid Systems - GA based BPNN	1

Reference Books

:COMPUTER SCIENCE AND ENGINEERING-CS2

1. Jyh-Shing Roger Jang, Chuen-Tsai Sun, Eiji Mizutani, Neuro-Fuzzy and Soft Computing, Prentice-Hall of India, 2003.
2. S.N. Sivanandam, S.N. Deepa, "Principles of Soft Computing", 2/e, John Wiley India, 2012.
3. James A. Freeman and David M. Skapura, Neural Networks Algorithms, Applications, and Programming Techniques, Pearson Edn., 2003.
4. Simon Haykin "Neural Networks and Learning Machines" Prentice Hall, 2008
5. Jon Krohn, Grant Beyleveld, Aglae Bassens, Deep Learning Illustrated, Pearson, 2020
6. Timothy J. Ross, "Fuzzy Logic with engineering applications", Wiley India, 2021
7. Bart Kosko.: Neural Network and Fuzzy Systems-Prentice Hall, Inc., Englewood Cliffs, 1991
8. Goldberg D.E.: Genetic Algorithms in Search, Optimization, and Machine Learning- Addison Wesley, 1989
9. Mitchell Melanie, An Introduction to Genetic Algorithm, Prentice Hall, 1998.

